

Pushkal Mishra

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Research Profile

My research focuses on scalable deep learning systems that perceive, reason, and act in the physical world. I work at the intersection of multimodal perception, sensor fusion, and vision-language-action learning, thereby connecting simulation, real-world sensing, and language grounding to support reliable autonomy.

Education

University of California San Diego

PhD, Electrical and Computer Engineering

Advisor: Prof. Dinesh Bharadia.

Aug 2024 - Present

San Diego, CA, USA

Indian Institute of Technology Hyderabad

B.Tech, Electrical Engineering; Computer Science minor

GPA: 9.33/10; rank: 4/62.

Nov 2020 - May 2024

Hyderabad, India

Industry Experience

AI Research Intern

Bosch Center for AI

Jun 2026 - Sep 2026

Pittsburgh, PA, USA

- Developed post-training framework for driving VLA models that uses simulator self-replay to learn from the consequences of alternative driving actions.
- Engineered parallel CARLA rollouts across GPU servers for scalable trajectory evaluation with simulated vehicle dynamics and traffic interactions.
- Optimized batched inference, KV caching, and simulation replay for scalable training-data collection, with drift controls to preserve fidelity.
- Implemented distributed LoRA fine-tuning to close the loop between simulation, post-training, and driving evaluation. Manuscript targeted for conference submission.

Research Projects

Radar-Language Scene Understanding

UC San Diego; advisor: Prof. Dinesh Bharadia

Jan 2025 - Present

San Diego, CA, USA

- Built a radar-language foundation model that supports scene description and object segmentation through a shared sensor representation.
- Introduced new spatially aware contrastive training to preserve object-location information, improving segmentation average precision by 21% relative to CLIP in simulation.
- Built a dataset and training pipeline for 800K+ radar-caption pairs from 110+ hours of simulation; trained the model in PyTorch across multiple A100 GPUs.
- Demonstrated sim-to-real transfer across radar sensors, adapting a lightweight decoder to real-world data while reusing the simulation-trained encoder.

Radar Simulation for Autonomous Driving

UC San Diego; advisor: Prof. Dinesh Bharadia

Aug 2024 - Feb 2025

San Diego, CA, USA

- Co-developed a physics-based radar simulator in CARLA that models how different materials reflect radar signals, enabling configurable sensor studies for autonomous driving.
- Built parallel data-generation pipelines in Kubernetes across towns, weather, and sensor configurations; evaluated whether radar-camera fusion could support end-to-end driving compared with LiDAR-camera baselines. [\[Project\]](#)

Memory-Efficient Neural Network Training

IIT Hyderabad; advisor: Prof. Ayon Borthakur

Jan 2023 - Jun 2024

Hyderabad, India

- Co-developed a neural-network training algorithm that learns in a single forward pass without backpropagation, reducing the memory required for training.
- Evaluated image classification and transfer learning in PyTorch; measured how training memory scales with network depth and batch size.

Graph Learning from Incomplete Sensor Data

IIT Hyderabad; advisor: Prof. Aditya Siripuram

Sep 2022 - Jan 2024

Hyderabad, India

- Developed a neural network that jointly reconstructs missing measurements and learns the graph connecting data sources, using an interpretable architecture derived from iterative optimization.
- Achieved graph-recovery F1 of 0.92 on synthetic data and evaluated robustness on temperature-sensor networks.

Publications

RLM: A Vision-Language Model Approach for Radar Scene Understanding

Pushkal Mishra, Kshitiz Bansal, Dinesh Bharadia.

arXiv:2511.21105, 2025; revised 2026. Manuscript under submission.

A Realistic Radar Simulator for End-to-End Autonomous Driving in CARLA

Satyam Srivastava*, Jerry Li*, **Pushkal Mishra***, Kshitiz Bansal, Dinesh Bharadia.

IEEE Vehicular Technology Conference (VTC Fall), 2025.

Learning Using a Single Forward Pass

Aditya Somasundaram*, **Pushkal Mishra***, Ayon Borthakur.

Transactions on Machine Learning Research (TMLR), 2025.

Inpainting-Driven Graph Learning via Explainable Neural Networks

Subbareddy Batreddy*, **Pushkal Mishra***, Yaswanth Kakarla, Aditya Siripuram.

IEEE Signal Processing Letters, vol. 32, pp. 111-115, 2025; Presented at ICASSP 2025.

Demo Abstract: C-Shenron: A Realistic Radar Simulation Framework for CARLA

Pushkal Mishra, Satyam Srivastava, Jerry Li, Kshitiz Bansal, Dinesh Bharadia.

ACM SenSys, 2025. Demo abstract.

* Equal contribution.

Skills

Programming: Python, C/C++, MATLAB, Bash.

Machine learning: PyTorch, Hugging Face, vision transformers, contrastive learning, vision-language-action models, LoRA fine-tuning, multi-GPU training, mixed precision.

Research systems: CARLA, Docker, Kubernetes, parallel simulation, model evaluation, batched inference, Linux, Git.